

The Relational Number System:

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Abstract

Prime numbers correspond to ratios in an alternative number system that is fundamentally more basic than our familiar set-based system. This document develops a relational number system in which prime numbers are defined as elementary, indivisible ratios or proportional transformations. By shifting the reference point from absolute quantities to pure relations, a system emerges that establishes multiplication as the primary operation and reflects the logarithmic structure of many natural laws.

.1 List of Symbols and Notation

Symbol	Meaning	Notes
Relational Basic Operations		
$\mathcal{P}_{\text{rel}1}$	Identity relation	1 : 1, starting point of all transformations
$\mathcal{P}_{\text{rel}2}$	Doubling relation	2 : 1, elementary scaling
$\mathcal{P}_{\text{rel}3}$	Fifth relation	3 : 2, musical fifth
$\mathcal{P}_{\text{rel}5}$	Third relation	5 : 4, musical major third
$\mathcal{P}_{\text{rel}p}$	Prime number relation	Elementary, indivisible proportion
Interval Representation		
I	Musical interval	As frequency ratio
$\vec{v}$	Exponent vector	$(a_1, a_2, a_3, \dots)$ for $2^{a_1} \cdot 3^{a_2} \cdot 5^{a_3} \dots$
p_i	i-th prime number	$p_1 = 2, p_2 = 3, p_3 = 5, p_4 = 7, \dots$
a_i	Exponent of i-th prime	Integer, can be negative
$n\text{-limit}$	Prime number limitation	System with primes up to n
Operations		
$\circ$	Composition of relations	Corresponds to multiplication
$\oplus$	Addition of exponent vectors	Logarithmic addition
$\log$	Logarithmic transformation	Multiplication $\rightarrow$ addition
$\exp$	Exponential function	Addition $\rightarrow$ multiplication
Transformations		
FFT	Fast Fourier Transform	Practical application
QFT	Quantum Fourier Transform	Quantum algorithm
Shor	Shor's Algorithm	Prime factorization

Table 1: Symbols and notation of the relational number system

.2 Introduction: Shifting the Reference Point

The idea of shifting the reference point to construct a number system based on ratios while reinterpreting the role of prime numbers is the key to a more fundamental understanding of mathematics. **Prime numbers correspond to ratios in an alternative number system that is fundamentally more basic** than our familiar set-based system.

What does shifting the reference point mean?

Previously, we have thought of the reference point (the denominator in a fraction like P/X) often as 1, representing a fixed, absolute unit. However, when we shift the reference point, we no longer think of absolute numerical values, but of **relational steps or transformations**.

Imagine we define numbers not as three apples, but as the **relationship or operation** that transforms one quantity into another.

.3 Music as a Model: Intervals as Operations

In music, an interval (e.g., a fifth, $3/2$) is not just a static ratio, but an **operation** that transforms one tone into another. When you shift a tone up by a fifth, you multiply its frequency by $3/2$.

Musical Intervals as a Ratio System

In just intonation, intervals are represented as ratios of whole numbers:

Interval	Ratio	Prime Factor	Vector
Octave	2 : 1	2^1	(1, 0, 0)
Fifth	3 : 2	$2^{-1} \cdot 3^1$	(-1, 1, 0)
Fourth	4 : 3	$2^2 \cdot 3^{-1}$	(2, -1, 0)
Major third	5 : 4	$2^{-2} \cdot 5^1$	(-2, 0, 1)
Minor third	6 : 5	$2^1 \cdot 3^1 \cdot 5^{-1}$	(1, 1, -1)

Table 2: Musical intervals in relational representation

These ratios can be written as **products of prime numbers with integer exponents**:

$$\text{Interval} = 2^a \cdot 3^b \cdot 5^c \cdot 7^d \cdot \dots \quad (1)$$

Depending on how many prime numbers one allows (2, 3, 5 – or also 7, 11, 13 ...), one speaks of a **5-limit**, **7-limit** or **13-limit** system.

Example .3.1 (A major third). The major third ($5/4$) can be expressed as $2^{-2} \cdot 5^1$:

$$\frac{5}{4} = 2^{-2} \cdot 5^1 \quad (2)$$

$$\text{Exponent vector: } (-2, 0, 1) \text{ for } (2, 3, 5) \quad (3)$$

Here this means:

- 2^{-2} : The prime number 2 appears twice in the denominator
- 5^{+1} : The prime number 5 appears once in the numerator

Vector Representation of Intervals

A useful representation is:

Definition .3.2 (Interval Vector).

$$I = (a_1, a_2, a_3, \dots) \text{ with } I = \prod_i p_i^{a_i} \quad (4)$$

Where:

- p_i : the i -th prime number (2, 3, 5, 7, ...)

- a_i : integer exponent (can be negative)

This allows a clear **algebraic structure** for intervals, including addition, inversion, etc. over the exponent vectors.

Application: Interval Multiplication = Exponent Addition

Example .3.3 (Major chord construction). Composition of the intervals of a C major chord in the 5-limit system:

$$C \rightarrow B = \mathcal{P}_{\text{rel}}1 \circ \text{Major third} \circ \text{Fifth} \quad (5)$$

$$= (0, 0, 0) \oplus (-2, 0, 1) \oplus (-1, 1, 0) \quad (6)$$

$$= (-3, 1, 1) \quad (7)$$

$$= \frac{2^{-3} \cdot 3^1 \cdot 5^1}{1} = \frac{15}{8} \quad (8)$$

The composition third $\circ$ fifth gives the major seventh C-B; the major triad C-E-G itself is the set $\{1, \frac{5}{4}, \frac{3}{2}\}$, third and fifth each measured from the root. This shows how complex harmonic structures emerge as compositions of elementary prime relations.

.4 Historical Precedents

The relational number system stands in a long tradition of mathematical-philosophical approaches:

- **Pythagorean harmony doctrine**: The Pythagoreans already recognized that *Everything is number* – understood as ratio, not as quantity
- **Euler's Tonnetz** (1739): Prime number-based representation of musical intervals in a two-dimensional lattice
- **Grassmann's Ausdehnungslehre** (1844): Multiplication as fundamental operation that creates new geometric objects
- **Dedekind cuts** (1872): Numbers as relations between rational sets

.5 Category-Theoretic Foundation

The relational system can be interpreted as a free monoidal category, where:

- **Objects** = ratio vectors $\vec{v} = (a_1, a_2, a_3, \dots)$
- **Morphisms** = proportional transformations between relations
- **Tensor product** $\otimes$ = composition $\circ$ of relations
- **Unit object** = identity relation $\mathcal{P}_{\text{rel}}1$

This structure makes explicit that the relational system has a natural category-theoretic interpretation.

.6 Prime Numbers as Elementary Relations

If we transfer this musical approach to numbers, we can interpret prime numbers not as independent numbers, but as **fundamental, irreducible proportional steps or transformations**:

The Elementary Ratios

Definition .6.1 (Prime Number Relations).

$\mathcal{P}_{\text{rel}}1$: Identity relation (1 : 1) (9)

The state of equality, starting point of all transformations (10)

$\mathcal{P}_{\text{rel}}2$: Doubling relation (2 : 1) (11)

The elementary gesture of doubling (12)

$\mathcal{P}_{\text{rel}}3$: Fifth relation (3 : 2) (13)

Fundamental proportional transformation (14)

$\mathcal{P}_{\text{rel}}5$: Third relation (5 : 4) (15)

Further elementary proportional transformation (16)

Numbers as Compositions of Ratios

In a relational system, numbers would not be static quantities, but **compositions of ratios**:

- **Starting point**: Base unit (1 : 1)
- **Numbers as paths**: Each number is a path of operations
 - The number 2: Path of the 2 : 1 operation
 - The number 3: Path of the 3 : 1 operation
 - The number 6: Path 2 : 1 followed by 3 : 1
 - The number 12: $2 \times 2 \times 3$ (three operations)

.7 Axiomatic Foundations

Axiom 1 (Relational Arithmetic). For all relations $\mathcal{P}_{\text{rel}}a, \mathcal{P}_{\text{rel}}b, \mathcal{P}_{\text{rel}}c$ in a relational number system:

1. **Associativity**: $(\mathcal{P}_{\text{rel}}a \circ \mathcal{P}_{\text{rel}}b) \circ \mathcal{P}_{\text{rel}}c = \mathcal{P}_{\text{rel}}a \circ (\mathcal{P}_{\text{rel}}b \circ \mathcal{P}_{\text{rel}}c)$
2. **Neutral element**: $\exists \mathcal{P}_{\text{rel}}1 \forall \mathcal{P}_{\text{rel}}a : \mathcal{P}_{\text{rel}}a \circ \mathcal{P}_{\text{rel}}1 = \mathcal{P}_{\text{rel}}a$
3. **Invertibility**: $\forall \mathcal{P}_{\text{rel}}a \exists \mathcal{P}_{\text{rel}}a^{-1} : \mathcal{P}_{\text{rel}}a \circ \mathcal{P}_{\text{rel}}a^{-1} = \mathcal{P}_{\text{rel}}1$
4. **Commutativity**: $\mathcal{P}_{\text{rel}}a \circ \mathcal{P}_{\text{rel}}b = \mathcal{P}_{\text{rel}}b \circ \mathcal{P}_{\text{rel}}a$

These axioms establish the relational system as an abelian group under the composition operation $\circ$.

.8 The Fundamental Difference: Addition vs. Multiplication

Addition: The Parts Continue to Exist

When we add, we essentially bring things together that exist side by side or sequentially. The original components remain preserved in some way:

- **Sets**: $2 + 3 = 5$ apples (original parts recognizable as subsets)
- **Wave superposition**: Frequencies f_1 and f_2 are still detectable in the spectrum
- **Forces**: Vector addition - both original forces are present

Multiplication: Something New Emerges

With multiplication, something fundamentally different happens. This involves scaling, transformation, or the creation of a new quality:

- **Area calculation:** $2m \times 3m = 6m^2$ (new dimension)
- **Proportional change:** Doubling $\circ$ tripling = sixfolding
- **Musical intervals:** Fifth $\times$ octave = new harmonic position

.9 The Power of the Logarithm: Multiplication Becomes Addition

The fact that taking logarithms turns multiplications into additions is fundamental:

$$\log(A \times B) = \log(A) + \log(B) \quad (17)$$

What does logarithmization teach us?

1. **Scale transformation:** From proportional to linear scale
2. **Nature of perception:** Many sensory perceptions are logarithmic
 - **Hearing:** Frequency ratios as equal steps
 - **Light:** Logarithmic brightness perception
 - **Sound:** Decibel scale
3. **Physical systems:** Exponential growth becomes linear
4. **Unification:** Addition and multiplication are connected by transformation

Logarithmic Perception

The nature of perception follows the Weber-Fechner law, which reflects the logarithmic structure of relational systems:

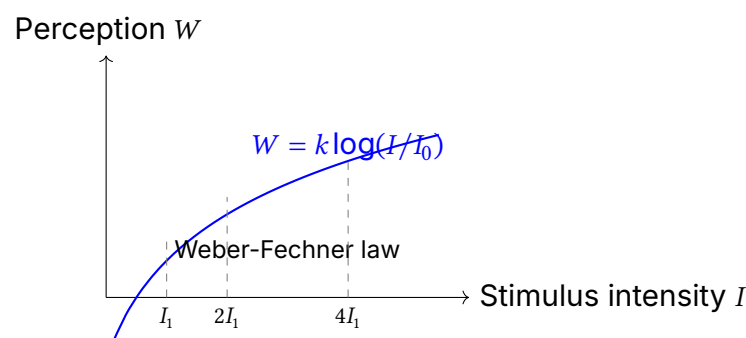


Figure 1: Logarithmic perception corresponds to the structure of relational systems

.10 Physical Analogies and Applications

Renormalization Group Flow

A remarkable parallel exists between relational composition and renormalization group flow in quantum field theory:

$$\beta(g) = \mu \frac{dg}{d\mu} = \sum_{k=1}^n \mathcal{P}_{\text{rel}} p_k \circ \log \left(\frac{E}{E_0} \right) \quad (18)$$

Here the energy scaling corresponds to the composition of prime relations.

Quantum Entanglement and Relations

Relational System	Quantum Mechanics
Prime relation $\mathcal{P}_{\text{rel}} p$	Basis state $ p\rangle$
Composition $\circ$	Tensor product $\otimes$
Vector addition $\oplus$	Superposition principle
Logarithmic structure	Phase relationships

Table 3: Structural analogies between relational and quantum systems

.11 Additive and Multiplicative Modulation in Nature

Electromagnetism and Physics

Modulation	Description	Examples
Multiplicative (AM)	Proportional amplitude change	Amplitude modulation, scaling
Additive (superposition)	Superposition of oscillations	Interference, beats

Table 4: Modulation in physics and technology

Music and Acoustics

- **Timbre:** Additive superposition of harmonic overtones with multiplicative frequency ratios
- **Harmony:** Consonance through simple multiplicative ratios (3 : 2, 5 : 4)
- **Melody:** Multiplicative frequency steps in additive time sequence

.12 The Elimination of Absolute Quantities

A central feature of this system is that the concrete assignment to a quantity is not necessary in the fundamental definitions. **The assignment to a specific quantity can be omitted and only becomes important when these relational numbers are applied to real things.**

Definition .12.1 (Relational vs. Absolute Numbers). • **Fundamental level:** Numbers are abstract relationships

- **Application level:** Measurement in concrete units (meters, kilograms, hertz)
- **Natural units:** $E = m$ (energy-mass identity as pure relation)

.13 FFT, QFT and Shor's Algorithm: Practical Applications

These algorithms already use the relational principle:

Fast Fourier Transform (FFT)

The FFT reduces complexity from $O(N^2)$ to $O(N \log N)$ through:

- Decomposition of the DFT matrix into sparsely populated factors
- Rader's algorithm for prime-sized transforms uses multiplicative groups
- Works with frequency ratios instead of absolute values

Quantum Fourier Transform (QFT)

- Quantum version of the classical DFT
- Core component of Shor's algorithm
- Works with exponential functions for period finding

Algorithmic Details: Shor's Algorithm

Algorithm 1 Shor's Algorithm for Prime Factorization

```

1: Input: Odd composite number  $N$ 
2: Output: Non-trivial factor of  $N$ 
3:
4: Choose random  $a$  with  $1 < a < N$  and  $\gcd(a, N) = 1$ 
5: Use quantum computer for period finding:
6:   Find period  $r$  of function  $f(x) = a^x \bmod N$ 
7:   Use QFT for efficient computation
8: if  $r$  is odd OR  $a^{r/2} \equiv -1 \pmod{N}$  then
9:   Go to step 4 (choose new  $a$ )
10: end if
11: Compute  $d_1 = \gcd(a^{r/2} - 1, N)$ 
12: Compute  $d_2 = \gcd(a^{r/2} + 1, N)$ 
13: if  $1 < d_1 < N$  then
14:   return  $d_1$ 
15: else if  $1 < d_2 < N$  then
16:   return  $d_2$ 
17: else
18:   Go to step 4
19: end if

```

The key lies in period finding through QFT, which recognizes relational patterns in modular arithmetic.

Algorithm	Property	Complexity	Application
FFT	Ratios	$O(N \log N)$	Signal processing
QFT	Superposition	Polynomial	Quantum algorithms
Shor	Period patterns	Polynomial	Cryptography

Table 5: Relational algorithms in practice

.14 Mathematical Framework

Formal Definition of the Relational System

Theorem .14.1 (Relational Number System). *A relational number system $\mathcal{R}$ is defined by:*

1. *A set of prime number relations $\{\mathcal{P}_{relP_1}, \mathcal{P}_{relP_2}, \dots\}$*
2. *A composition operation $\circ$ (corresponds to multiplication)*

3. A vector representation $\vec{v} = (a_1, a_2, \dots)$ with $\prod_i p_i^{a_i}$
4. A logarithmic addition operation $\oplus$ on vectors

Properties of the System

- **Closure:** $\mathcal{P}_{\text{rel}}a \circ \mathcal{P}_{\text{rel}}b \in \mathcal{R}$
- **Associativity:** $(\mathcal{P}_{\text{rel}}a \circ \mathcal{P}_{\text{rel}}b) \circ \mathcal{P}_{\text{rel}}c = \mathcal{P}_{\text{rel}}a \circ (\mathcal{P}_{\text{rel}}b \circ \mathcal{P}_{\text{rel}}c)$
- **Identity:** $\mathcal{P}_{\text{rel}}1$ is neutral element
- **Inverses:** Each relation $\mathcal{P}_{\text{rel}}a$ has inverse $\mathcal{P}_{\text{rel}}a^{-1}$

.15 Advantages and Challenges

Advantages of the Relational System

1. **Fundamental nature:** Captures the essence of relationships
2. **Logarithmic harmony:** Compatible with natural laws
3. **Multiplicative primary operation:** Natural connection
4. **Practical application:** Already implemented in FFT/QFT/Shor

Challenges

1. **Addition:** Complex definition in purely relational spaces
2. **Intuition:** Unfamiliar for set-based thinking
3. **Practical implementation:** Requires new mathematical tools

.16 Epistemological Implications

The relational number system has profound philosophical consequences:

- **Operationalism:** Numbers are defined by their transformative effects, not by static properties
- **Process ontology:** Being is understood as a dynamic network of transformations
- **Neo-Pythagoreanism:** Mathematical relations as fundamental substrate of reality
- **Structuralism:** The structure of relationships is primary over *objects*

.17 Open Research Questions

The relational number system opens various research directions:

1. **Canonical addition:** How can addition be naturally defined in the relational system without transitioning to logarithmic space?
2. **Topological structure:** Is there a natural topology on the space of prime relations?
3. **Non-commutative generalizations:** Can the system capture quantum groups and non-commutative structures?
4. **Algorithmic complexity:** Which computational problems become easier or harder in the relational system?

5. **Cognitive modeling:** How is relational thinking reflected in neural structures?

.18 Appendix A: On the T0 Factorization Tool

A T0 factorization tool that was meant to apply the relational principle to prime factorization does not support the statements attributed to it, because there is no T0 or ξ factorization method distinguishable from Shor; the method claim is withdrawn. The adaptive scaling of ξ with problem size is a hard-wired case distinction by bit length whose branches behave identically, and is therefore no evidence of a dependence of ξ on the number range. The resonance function $\exp(-(\omega - \pi)^2/(4|\xi|))$ with $\omega = 2\pi/r$ has its maximum at $r = 2$ for every ξ ; it selects the smallest even order, and the period search remains a classical order search.

The gates of the tool carry no physical statement either: with the factor correction $= 1 + \xi$ the Hadamard and CNOT gates are not unitary and the norm is not preserved: $P(00) + P(11) = (1 + \xi)^2/2 + (1 + \xi)^4/2$, i.e. $1.0004 \neq 1$ for $\xi = 4/30000$. A Bell state therefore results only after renormalisation.

Practical applications of the relational principle are the standard methods named in the section on FFT, QFT and Shor's algorithm.